ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PRACTICE EXAM 6 (PAPER 2)

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Practical paper in the style of the DHET second papers: design the breadboard circuit and flowchart of a box-sorting cell, program it on an Arduino, design and slice a 3D-printed sensor bracket, and design an IoT link that sends the cell's counts to a cloud dashboard.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. This is a practical paper. Use Fritzing for circuits, draw.io for flowcharts and block diagrams, the Arduino IDE for programs, and a CAD program (such as Tinkercad or Fusion 360) and a slicer (such as Cura) for QUESTION 3. If you do not have a computer, draw and write your answers neatly by hand.
3. Put the question number and your name or examination number on every printout or page. Save your work regularly.
4. Programs are not marked strictly on syntax.

QUESTION 1

> A packaging plant wants a small prototype cell that sorts boxes by height. Boxes travel on a conveyor driven by a DC motor. When an IR sensor detects a box at the measuring point, the conveyor stops and an ultrasonic sensor mounted 30 cm above the belt measures the box. Boxes taller than 15 cm are pushed into the "tall" lane by a servo-driven diverter arm; shorter boxes go straight on. The conveyor then starts again.

Components:

- Arduino Uno
- IR obstacle sensor module (OUT goes LOW when a box is present)
- HC-SR04 ultrasonic sensor
- SG90 servo motor (diverter arm)
- DC motor with an L298N motor driver (conveyor), and a 9 V supply for the motor
- Green LED with a 220 Ω resistor (conveyor running)
- Breadboard and jumper wires

1.1 Using the components above, design the breadboard circuit of the sorting cell in Fritzing. Show the separate motor supply with a common ground. (14)

1.2 Draw a flowchart of the logic of the sorting cell, including the measurement, the decision, the diverter action and restarting the conveyor. (14)

1.3 Save and print (or neatly present) the circuit and the flowchart. (2)

[30]

QUESTION 2

Use these connections: IR sensor OUT to pin 2; HC-SR04 Trig to pin 7 and Echo to pin 8; servo to pin 9; L298N ENA to pin 5, IN1 to pin 6 and IN2 to pin 4; green LED to pin 12.

2.1 Write an Arduino program for the sorting cell that does the following: (28)

• Sets up all pins, the servo (diverter at 0°) and the serial monitor, and starts the conveyor at a PWM value of 180 with the green LED on.

• When the IR sensor detects a box, stops the conveyor and switches the LED off.

• Measures the distance to the top of the box and calculates the box height (height = 30 − distance).

• If the height is more than 15 cm, moves the diverter to 90°, waits 1 second and returns it to 0°, and adds 1 to a tall-box counter; otherwise adds 1 to a short-box counter.

• Prints both counters on the serial monitor, waits until the box has left the sensor, and restarts the conveyor.

• Prints "Measurement error" and does not move the diverter if the distance is 0 or more than 30 cm.

• Is well commented.

2.2 Save and print (or neatly present) the program. (2)

[30]

QUESTION 3

3.1 Using a CAD program, design a bracket that holds the HC-SR04 above the conveyor. The bracket is a flat plate of 50 mm × 25 mm × 3 mm with two 16 mm holes for the sensor's transducers, 26 mm apart (centre to centre), and a 90° flange at the top edge of 50 mm × 15 mm × 3 mm with two 4 mm holes for screws. Produce a dimensioned drawing (or screenshots) of your design. (15)

3.2 Export your design as an STL file and prepare it in a slicer. Copy and complete the table below with the settings you chose and give a reason for EACH setting. (10)

Setting	Value chosen	Reason
Filament material		
Layer height		
Infill percentage		
Supports (yes / no)		
Print orientation		

3.3 The plant manager wants to see the box counts on a cloud dashboard. Draw a block diagram (in draw.io) that shows how an ESP32 board could read the counts and send them over Wi-Fi to an MQTT broker and a cloud dashboard. Label every block and the direction of the data, and name the protocol on each link. (12)

3.4 Save and print (or neatly present) your bracket drawing, slicer table and block diagram. (3)

[40]

TOTAL: 100

END OF QUESTION PAPER